

Stellar PSFs Fake Hosts in Little Red Dots

Broad-band point-spread functions built from field stars misassign light when source spectra differ.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for compact high-redshift nuclei

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Background

Little red dots are compact early-Universe sources whose unresolved nuclei and possible hosts overlap in the same diffraction-limited image. Their structure is usually inferred by subtracting a point-spread function modelled from nearby field stars.

That stellar PSF assumes the target's spectrum matches the stars, an assumption never tested for spectrally distinct compact sources.

What this paper is selling

The authors derive why broad-band stellar PSFs systematically misassign flux for sources with different spectral slopes, then demonstrate a source-conditioned polychromatic correction. The effect is large enough to change inferred sizes, luminosities and masses of little red dots.

Key results to remember

- A pure point source with no host produces wavelength-dependent residuals mimicking a Sérsic component.
- Every tested nuclear spectrum, stellar reference and detector position exceeds the published 11.8% F444W host fraction.
- The bias inflates or fabricates extended emission, altering sizes, luminosities and masses inferred for compact sources.

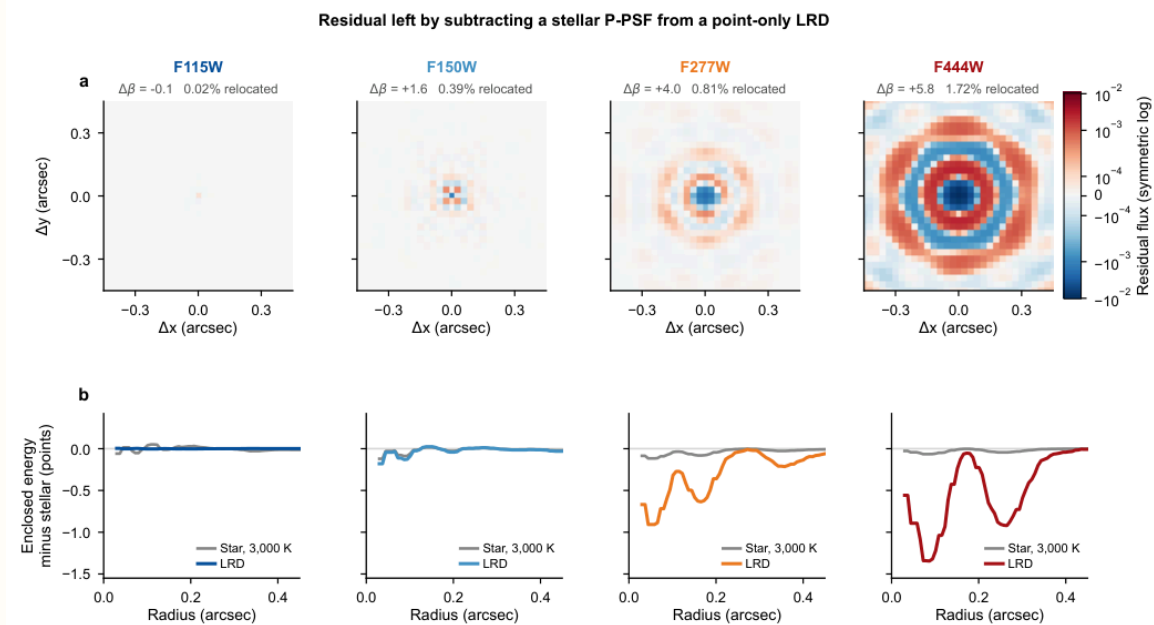


Figure 1 | Subtracting a stellar PSF from a point-only LRD leaves filter-dependent residuals that masquerade as a Sérsic host.

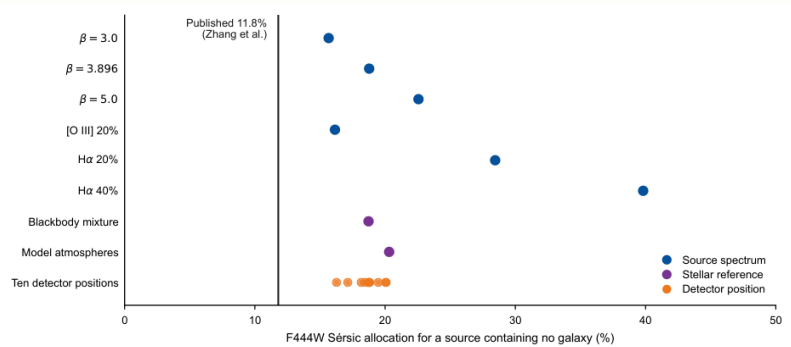


Figure 2 | Across all tested spectra, references and detector positions, the spurious F444W host allocation always exceeds the published 11.8% value.